

Modeling CRISPR perturbations in melanoma cells under different experimental conditions

For this project, you are provided with a CRISPR perturbation dataset by [Frangieh et al., 2021](#). It contains scRNA-seq data of ca. 218,000 melanoma cells split between three experimental conditions and treated with a CRISPR library targeting 248 genes. You also have access to the single-cell surface protein measurements, which you may use for task 3 (see below). More information on the dataset can be found in the publication.

You will be working in groups of 1-3 students. The scope of the project is adjusted to the group size as marked in the tasks below.

Task 1: Condition Classification

Can a classifier identify which treatment condition a cell came from using its gene expression profile? Which genes drive the separation?

Train one or more classification models (logistic regression, tree ensembles or neural network), depending on group size:

- a) One student: Choose one type of model,
- b) Groups with two students: Choose two types of models.
- c) Groups with three students: Train one of each.

first look at data, look at quality ->
do more QC/preprocessing if you
want

For each model you train, compute feature importance and discuss the results.

Task 2: Clustering

Which genetic perturbations show similar effects, if any? Which clustering method(s) best capture(s) the underlying biology?

Apply different clustering methods to the data, visualize and compare the results. Students working alone only need to use data from the co-culture condition. Interpret the results biologically. Do the clusters correspond to what you would expect based on the findings described in the paper, e.g., the pathways that are discussed?

Task 3: Perturbation Prediction

Can a model predict transcriptome changes for a gene that was not knocked out in the training data?

Before model training, select a subset of 50 perturbations for modeling and explain your choice. Of these, select a subset of perturbations that are held out, i.e., not used during training or hyperparameter selection, only at test time. If the number of selected genes is still too high for the model you want to train, you may subset it further and explain your choice. You may also use unperturbed cells.

The target variable to predict by your models is the mean $\log_2$ fold change per condition and perturbation target. Train two (if one student), three (if two students) or four (if three

students) types of models. Here, a type of model is a feature engineering strategy combined with a learning algorithm. At least one of the models should be deliberately simplistic. Evaluate the model performance and uncertainty based on suitable metrics and interpret the results biologically.

You may also use the protein data, however, this is not required. Students working alone only need to use data from the co-culture condition.

Dates

Code submission: **August 28**, 23:59

Presentation:

- Earliest: 2 weeks after code submission
- Latest: October 2

Please indicate whether you will work on your own or who you team up with in [this document](#) by the end of **July 03**.

Grades will be assigned after all groups have presented their projects.

Grading criteria

The course will be graded based on your submission, as well as the presentation and Q&A. Note that, to be awarded the full points, it is sufficient to thoroughly complete the tasks above. Should you want to include additional analyses, you might receive additional points for them.

Code submission

Please hand in scripts and/or notebooks that are well structured and contain code comments or annotations. For each task, also include a discussion of the results including the biological context (ca. 0.5-1 pages each).

Experimental setup

Design your experiment appropriately, e.g., using one of the data splitting strategies presented in Lecture 4. Justify your choices.

Evaluate the influence of relevant model parameters and justify your choices. For example, for k -means clustering, you might set $k = 3$ and mention that you expect the data to be separable into three classes based on your previous PCA plot, or you might train several models with different k and evaluate which one is best for the task using suitable metrics.

Model evaluation

We expect you to design your models carefully to improve their performance. However, for your grade, how well the model actually performs is less important than a sensible and thorough evaluation of the model's performance, using suitable metrics.

For tasks 1 and 3: If multiple models are trained, compare results of your models, including in the context of the bias-variance tradeoff. Give a hypothesis describing the differences you see between models.

Presentation and Q&A

Please prepare a presentation of 15 minutes, which will be followed by 15 minutes of Q&A. In your presentation, give a very brief overview of the data and biological context (1 slide), then walk us through your solutions of the tasks. Focus on explaining and justifying rather than merely describing your approach and the results.

For projects submitted by groups, please allocate some presentation time to each student. We also expect every student of the group to be able to answer questions about experimental design and model evaluation of the entire submission, though we are aware that familiarity with the code will be highest for the student who wrote it. We may also ask you to explain excerpts of your code submission.

Usage of AI

Usage of AI is allowed for generating code, but we discourage excessive use, as it inhibits learning how to plan experiments and code.

Do not use AI to generate your discussions of results. You may use it to polish text, e.g., for spelling, grammar, and typographical issues.

Note that during the Q&A after your presentation, we may ask you any questions about the code and discussions of results you submit. The answers will be graded, so you should fully understand any submitted materials.